

A Comparative Study of Point-Cloud Segmentation Methods for Walkable Tread Detection on Staircases under Sensor Noise

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Abstract—Stair-climbing robots require reliable three-dimensional perception to distinguish walkable surfaces from hazardous structural parts. This work builds a segmentation pipeline for a synthetic point cloud of a four-step staircase generated from mathematical equations together with LiDAR noise and outliers. Four segmentation methods are implemented and compared, namely RANSAC plane fitting, PCA-based normal vector analysis, DBSCAN clustering with slope analysis, and height histogram analysis. The main contribution is a controlled, fully labeled synthetic benchmark together with a unified comparison of these methods and a systematic sensor-noise sensitivity analysis oriented toward robot safety. Evaluation is performed on the walkable class using accuracy, precision, recall, F1, and IoU. The results show that the height histogram and the normal analysis methods achieve the best performance, with F1 scores of 0.883 and 0.876 respectively and precision above 0.93. A sensitivity analysis shows that performance degrades monotonically as noise increases, with F1 dropping from 0.970 under noise-free conditions to 0.329 at a noise standard deviation of 0.08 meters. The safety discussion concludes that the segmentation results are adequate as an initial perception layer, but must be complemented by geometric safety margins and sensor fusion before autonomous use by a robot.

Index Terms—point cloud, segmentation, RANSAC, normal vector, DBSCAN, stair detection, robot

I. INTRODUCTION

The ability of a mobile robot to climb stairs is an important challenge in mobile robotics because staircases are common structures in human-made environments. A stair-climbing robot must accurately identify the horizontal walkable surfaces (treads) and distinguish them from vertical surfaces (risers) as well as the hazardous edges of the staircase. An error in identifying these surfaces can cause the robot to slip or fall, so segmentation accuracy becomes a critical safety factor.

LiDAR sensors and depth cameras produce a representation of the environment in the form of a three-dimensional point cloud. Processing point clouds to detect geometric structures such as staircases requires segmentation techniques capable of separating points based on surface orientation and height. Several classical approaches have been developed, including RANSAC-based plane fitting [1], [2], surface normal estimation [5], [6], and density-based clustering algorithms [3], [4].

Building on these primitives, stair detection from 3D point clouds and depth cameras has been studied for both assistive devices and mobile robots [9]–[12].

This study aims to: first, generate a simulated point cloud of a four-step staircase using mathematical equations that represent the staircase geometry together with sensor noise; second, implement and compare several segmentation algorithms to distinguish walkable surfaces, vertical surfaces, and other parts; third, quantitatively evaluate the performance of each method and analyze the effect of noise; and fourth, discuss the suitability of the segmentation results for deployment on a stair-climbing robot.

The contributions of this work are threefold: (i) a controlled and fully labeled synthetic staircase point-cloud benchmark with adjustable noise and outliers; (ii) a unified, reproducible comparison of four classical segmentation methods under identical conditions; and (iii) a sensor-noise sensitivity study together with a safety-oriented analysis of the walkable-class precision required by stair-climbing robots.

II. POINT CLOUD GENERATION

This subsection discusses the generation of the synthetic staircase point cloud. The staircase is modeled with four steps using the geometric parameters in Table I. Each point is assigned a ground-truth label in one of three classes, namely the tread class (walkable), the riser class (vertical), and an other class that covers outliers and noise.

TABLE I
STAIRCASE GEOMETRIC PARAMETERS

Parameter	Symbol	Value
Staircase width (x-axis)	w	2.5 m
Tread depth (y-axis)	d	0.30 m
Step height (z-axis)	h	0.18 m
Number of steps	n	4
Vertical noise std. deviation	σ_z	0.02 m
LiDAR noise std. deviation	σ_L	0.005 m
Outlier fraction	—	4 %

A. Horizontal Surface (Tread)

The tread is a horizontal surface on which the robot can step. For each step i with $i = 0, 1, \dots, 3$, points are generated using (1).

$$\begin{cases} x \sim \text{Uniform}(0, w) \\ y \sim \text{Uniform}(i d, (i+1) d) \\ z = i h + \epsilon_z, \quad \epsilon_z \sim \mathcal{N}(0, \sigma_z^2) \end{cases} \quad (1)$$

The noise term ϵ_z represents the uncertainty in the sensor height measurement, so that the tread surface is not perfectly flat.

B. Vertical Surface (Riser)

The riser is a vertical surface connecting two consecutive treads. For each step i with $i = 1, 2, 3$, points are generated using (2).

$$\begin{cases} x \sim \text{Uniform}(0, w) \\ y = i d + \epsilon_y, \quad \epsilon_y \sim \mathcal{N}(0, \sigma_y^2) \\ z \sim \text{Uniform}((i-1) h, i h) \end{cases} \quad (2)$$

The riser for $i = 0$ is not generated because the equation would yield z below the floor, which is inconsistent with the geometry of an ascending staircase.

C. LiDAR Noise and Outliers

After all points are generated, isotropic noise $\mathcal{N}(0, \sigma_L^2)$ is added on all three axes to mimic LiDAR sensor noise. Outliers amounting to four percent of the total points are then added, scattered randomly throughout the bounding box. This configuration yields a total of 15,808 points, comprising 10,400 tread points, 4,800 riser points, and 608 other points. A visualization of the raw point cloud is shown in Fig. 1.

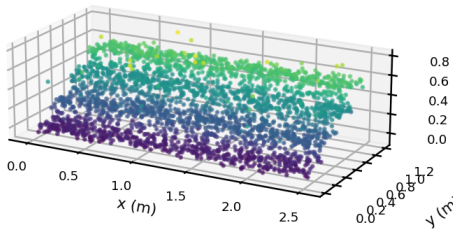


Fig. 1. Visualization of the raw point cloud of the four-step staircase. Color indicates height z ; the four steps are clearly visible along with the scattered outliers.

III. SEGMENTATION METHODOLOGY

This section discusses the four implemented segmentation methods. Most of the methods rely on local normal vector estimation, which is described first.

A. Normal Vector Estimation

The normal vector at each point is estimated through Principal Component Analysis (PCA) over the k nearest neighbors, a standard local-surface estimator widely used in point-cloud processing [5]–[7]. The local covariance matrix is defined in (3).

$$\mathbf{C}_p = \frac{1}{k} \sum_{j=1}^k (\mathbf{q}_j - \bar{\mathbf{q}})(\mathbf{q}_j - \bar{\mathbf{q}})^\top \quad (3)$$

where $\mathbf{q}_j$ is a neighboring point and $\bar{\mathbf{q}}$ is its mean. The normal vector $\mathbf{n}_p$ is obtained from the eigenvector corresponding to the smallest eigenvalue of $\mathbf{C}_p$, as in (4).

$$\mathbf{n}_p = \arg \min_{\|\mathbf{v}\|=1} \mathbf{v}^\top \mathbf{C}_p \mathbf{v} \quad (4)$$

The vertical component $|n_z|$ is used as an orientation indicator: a value close to one denotes a horizontal surface, whereas a value close to zero denotes a vertical surface. In these experiments, $k = 30$ is used.

B. Method 1: RANSAC Plane Fitting

The RANSAC method [1], [2] is used to detect horizontal planes whose normal vector is close to the z -axis. At each iteration, three random points are selected to define a candidate plane $\mathbf{n}^\top \mathbf{x} + d = 0$. A candidate is accepted only if $|n_z| > 0.9$, after which the number of inliers is counted from points lying within a threshold $\tau = 0.02$ m of the plane. The plane with the most inliers is extracted iteratively using an inlier-removal strategy. Because a horizontal plane band can also capture riser points that happen to lie at the same height, plane membership is combined with a local normal consistency constraint ($|n_z| > 0.8$). This combination emphasizes that pure geometric plane fitting needs to be supported by normal analysis.

C. Method 2: PCA-Based Normal Vector Analysis

This method classifies each point directly based on the vertical component of its normal vector, as in (5).

$$\text{label}(p) = \begin{cases} \text{tread} & |n_z| > \tau_h \\ \text{riser} & |n_z| < \tau_v \\ \text{other} & \text{otherwise} \end{cases} \quad (5)$$

with thresholds $\tau_h = 0.85$ and $\tau_v = 0.35$. Points whose normal orientation lies between the two thresholds are treated as edges or transitions.

D. Method 3: DBSCAN Clustering with Slope Analysis

In a dense staircase, the tread and riser surfaces touch each other, so pure spatial clustering tends to merge them into a single cluster. Therefore, the DBSCAN algorithm [3], [4] is used primarily as an outlier rejector with parameters $\varepsilon = 0.04$ m and a minimum sample count of ten. Points marked as noise are assigned the other class, while valid points are classified using local normal slope analysis as in the second method. This combination of density clustering and slope analysis provides robustness against outliers.

E. Method 4: Height Histogram Analysis

The tread surfaces are concentrated at discrete heights $z \approx 0, h, 2h, 3h$. A histogram of the z values forms a peak at each level. The n tallest peaks that are separated by more than eight centimeters are selected as tread levels. Points lying within a tolerance of ± 0.04 m of a peak and having a vertical normal ($|n_z| > 0.75$) are marked as tread. The detected peaks are 0.009, 0.181, 0.352, and 0.542 meters, which are very close to the ideal multiples of the step height.

IV. EXPERIMENTAL RESULTS

This section discusses the segmentation results and the quantitative evaluation. Performance is measured on the walkable class, with tread as the positive class. Based on the number of true positives (TP), false positives (FP), false negatives (FN), and true negatives (TN), the metrics are computed using (6).

$$\begin{aligned} \text{Acc} &= \frac{TP + TN}{N}, & \text{Prec} &= \frac{TP}{TP + FP}, & \text{Rec} &= \frac{TP}{TP + FN} \\ F1 &= \frac{2 \text{Prec} \cdot \text{Rec}}{\text{Prec} + \text{Rec}}, & \text{IoU} &= \frac{TP}{TP + FP + FN} \end{aligned} \quad (6)$$

TABLE II
SEGMENTATION PERFORMANCE COMPARISON FOR THE WALKABLE CLASS

Method	Acc	Prec	Rec	F1	IoU
RANSAC + Normal	0.836	0.947	0.795	0.864	0.761
PCA + Normal	0.848	0.944	0.816	0.876	0.779
DBSCAN + Slope	0.843	0.957	0.798	0.870	0.770
Height Histogram	0.854	0.935	0.836	0.883	0.790

Table II summarizes the performance of the four methods. The height histogram method achieves the highest F1 and IoU, while DBSCAN gives the highest precision. All methods reach a precision above 0.93, which means that the parts marked as walkable are almost always genuinely treads. A visual comparison of the metrics is presented in Fig. 2.

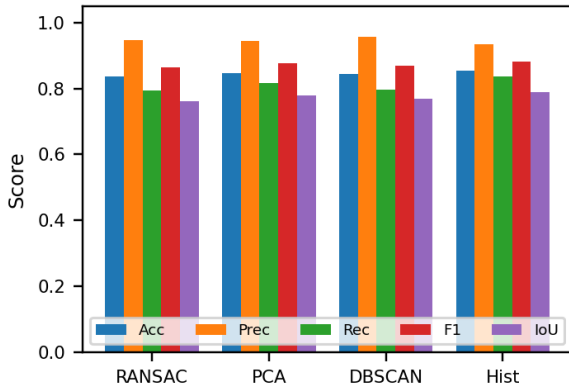


Fig. 2. Metric comparison across methods for the walkable class.

The height distribution together with the detected tread peaks is shown in Fig. 3. Four clear peaks confirm that the

histogram approach effectively separates the staircase levels. The three-dimensional segmentation result compared with the ground-truth labels is shown in Fig. 4.

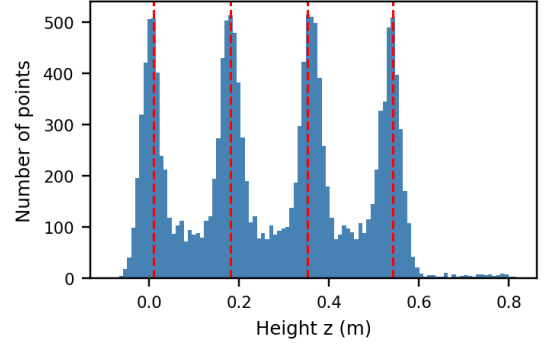


Fig. 3. Histogram of height z . The dashed red lines mark the detected tread levels.

V. ANALYSIS AND DISCUSSION

A. Effect of Noise on Accuracy

This subsection discusses the sensitivity of performance to noise. The vertical noise standard deviation σ_z is varied from zero to 0.08 meters, after which the normal analysis pipeline is rerun. The result in Fig. 5 shows a monotonic decrease in performance. The F1 score drops from 0.970 under noise-free conditions to 0.875 at $\sigma_z = 0.02$ m, then falls sharply to 0.551 at $\sigma_z = 0.04$ m and to only 0.329 at $\sigma_z = 0.08$ m. This decline occurs because large noise corrupts the local normal vector estimation, so the orientation thresholds can no longer reliably separate treads and risers.

B. Suitability for Stair-Climbing Robots

This subsection discusses the implications of the results for robot safety. In the context of a stair-climbing robot, precision is more critical than recall because a false positive, that is, a riser or edge mistakenly marked as walkable, can cause the robot to slip. The precision of about 0.94 achieved by the four methods is good, but for real deployment it should be raised beyond 0.99 through edge filtering, morphological erosion of the tread area, and the addition of a geometric safety margin.

In addition, the sensitivity to noise demands the use of high-precision sensors or temporal filtering over multiple scans, as adopted in odometry-aided stair traversal systems [11]. The $|n_z|$ threshold should be adaptive to the estimated noise level. The staircase edge is the most hazardous yet the most difficult region to segment because its normal vector is transitional, so it is advisable to define a dedicated edge class together with a no-step zone.

Based on the overall analysis, these segmentation results are adequate as an initial perception layer, but are not yet sufficient as the sole safety guarantee. Safe deployment requires sensor fusion with an inertial measurement unit and a camera, temporal verification, geometric safety margins, and a contact-based fail-safe mechanism.

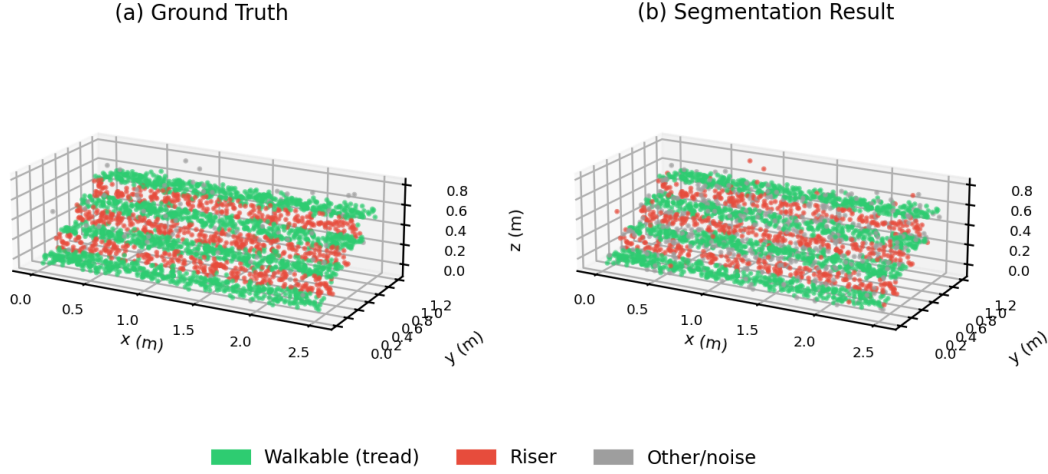


Fig. 4. Comparison of segmentation results. (a) ground-truth labels; (b) result of the height histogram method. Green denotes walkable surfaces, red denotes risers, and gray denotes the other class.

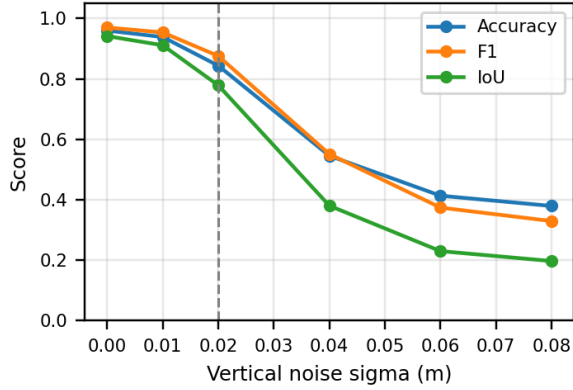


Fig. 5. Effect of vertical noise on accuracy, F1, and IoU. The dashed line marks the noise level used in the main experiment.

VI. CONCLUSION

This study successfully built a complete pipeline for walkable surface segmentation on a four-step staircase point cloud. The synthetic data were generated from mathematical equations together with LiDAR noise and outliers, yielding more than fifteen thousand labeled points. Four segmentation methods were implemented and compared under identical conditions, namely RANSAC plane fitting, PCA-based normal analysis, DBSCAN with slope analysis, and height histogram analysis. The height histogram and normal analysis methods provided the best performance, with F1 close to 0.88 and IoU close to 0.79. The sensitivity analysis proved that performance decreases monotonically with noise, underscoring the importance of sensor quality and preprocessing. Future work is directed toward defining an edge class, noise-adaptive thresholds, smoothness-based region growing [8], learning-based detectors [12], and validation on real point cloud data.

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